VENUSGEO & PRIVATE IDENTITY®

Ultrapass® for Carnival

Standards-Based Identity for Guests, Crew, and the Connected Ship

Carnival DigitalID (Guests) • Carnival CrewID (Workforce)

Powered by the Ultrapass Authenticator SDK | NIST 800-63 IAL2 / AAL2 | FIDO2 Passkeys | On-Device Biometrics

Prepared for Carnival Corporation Executive, Engineering, and Product Leadership

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1. Executive Summary

Carnival closed FY2025 with record revenue of $36.4 billion, roughly 225,000 guests aboard on any given day, and over 100,000 shipboard employees. At that scale, **every second of friction at check-in, the bar, the casino entrance, the crew tablet, and the manager's approval is a measurable cost — and every second removed is measurable revenue and capacity.**

VenusGeo proposes the PrivateID Ultrapass SDK product: **Carnival DigitalID** for guests — verify at home, carry the industry's first digital identity wallet, approve purchases with a selfie, prove age in real time — and **Carnival DigitalID "CrewID"** for the workforce — passwordless face + passkey login on shared ship devices, manager selfie approvals, and Bluetooth muster accountability.

The architectural difference from Carnival's incumbent Thales-based facial-recognition stack comes down to one principle: **Ultrapass sends the answer across the network — never the biometric.** All capture, liveness, matching, and age estimation execute on the guest's or crew member's own device. What crosses the network is a signed, kilobyte-scale verdict — "this is the enrolled guest," "over 18 or 21," "present at muster station 4", "crew member is logged into device #186" — delivered into Carnival's existing applications as standard OIDC claims or SAML attributes. No facial image or biometric template ever leaves the device. There is no central biometric database to build, secure, or breach; verification keeps working when satellite links don't; and coverage scales to every phone and tablet the fleet already owns, at near-zero marginal cost per touchpoint.

Ultrapass is additive from day one. Carnival's existing voyage facial-recognition capability and the CBP Traveler Verification Service debarkation partnership continue untouched. Integration lands in approximately two working weeks per application — pre-cruise Online Check-In, the Carnival HUB app, TmHub, Funtime — with no middleware server to procure or operate. The engine is production-proven at roughly 3.5 million on-device age estimations per month for large global search and social media platforms, measured in NIST's FRTE 1:1 verification track, and built end to end on open standards — which means Carnival's applications decouple from any vendor's matcher SDK permanently. **This is the last identity migration of its kind.**

An illustrative financial model (Section 8) shows a base-case annual benefit of $75–150M+ against a software-only cost structure — onboard revenue lift, terminal throughput, workforce time recapture, and fraud/chargeback reduction — with payback measured in months because there is no hardware denominator.

Why now

- **Demand is at all-time highs.** Carnival is ~90% booked for the remainder of 2026 at historically high prices. The revenue constraint is increasingly throughput and onboard conversion — exactly what identity friction throttles.
- **Biometrics are proven in this industry — at the gangway.** CBP facial comparison cut cruise debarkation clearance up to 30%, verifying in ~2 seconds at >98% accuracy; biometric embarkation at Vancouver cut per-passenger processing from ~2–3 minutes to ~10 seconds and total embarkation from ~2 hours to ~30 minutes. Guests now expect that experience everywhere on the ship, not just at its edge.
- **The regulatory wind favors on-device.** In March 2026 Spain's data protection authority fined an age-assurance provider €950,000 for server-side biometric processing, invalid consent, and excessive retention (decision under appeal). Every one of those findings attaches to central biometric processing. An architecture that never centralizes a biometric cannot over-retain one.
- **First-mover credential advantage.** No cruise line has issued its guests a portable, standards-based digital identity wallet. **Carnival would be first** — and a verified, consented identity wallet is a durable loyalty and data asset across all nine brands, not just a security control.

2. Identity Is a P&L Line at Carnival's Scale

Carnival Cruise Line operates 27+ ships, carrying roughly 18 million passengers a year, and generates roughly one-third of total revenue — about $8+ billion in FY2025 — from onboard and other spend. Onboard spend is transaction-dense: bars, casinos, specialty dining, retail, spa, shore excursions, and adult-only venues, most of which require an identity or age check that today runs on cards, PINs, wristbands, manual document inspection, or shared credentials.

Every one of those checks is an identity event, and across the fleet they number in the hundreds of millions per year. Small per-event improvements compound into material financial outcomes:

Friction point today	Cost mechanism	Ultrapass effect
Terminal check-in & document inspection	Labor hours, queue time, delayed first onboard purchase, guest NPS	Enroll at home to IAL2; verified guests walk on with a QR scan or face check
Age checks at bars, casino, adult venues	Slow service, compliance exposure, abandoned transactions	Real-time age proof from the credential or on-device facial age estimation — no document handling
Crew logins on shared devices (PINs, cards, passwords)	Minutes lost per shift × 100,000 crew; credential sharing destroys accountability	Sub-second face + passkey login; every session bound to a verified individual
Manager approvals (comps, refunds, voids)	Managers walk to terminals; weak audit trails invite shrinkage	Selfie approval on any nearby tablet with a cryptographic audit log
Family purchases & minors' activities	Revenue left on the table when parents can't easily authorize teen spend	Family accounts with parental selfie sign-off unlock supervised spend
Muster & emergency accountability	Minutes of radio roll-call; manual reconciliation under pressure	Signed Bluetooth presence events roll up to a live accountability view on the bridge

Identity is not an IT cost center. At Carnival's scale it is a throughput, revenue, and labor-productivity lever.

3. What Carnival Runs Today — and What It Can't Reach

Carnival's biometric estate today has two layers, and this proposal touches neither. **At U.S. terminals,** debarkation runs on CBP's Traveler Verification Service — government-operated infrastructure whose vendor choices belong to CBP, not Carnival. **Onboard,** Carnival operates a voyage-scoped facial-recognition capability — confirmed by Carnival's team as Thales-based — for gangway identification and for associating onboard photographs with passengers in the HUB app, with biometric identifiers deleted after the voyage. These flows work, and Ultrapass runs beside them from day one.

The gap is everything else: the hundreds of daily identity moments per ship that happen after the gangway, plus the entire pre-cruise journey. The incumbent model — server-side matching against a central gallery, dedicated capture cameras, GPU infrastructure, ship-to-shore connectivity — is architecturally a **hardware-and-server-anchored system.** It cannot economically extend to a bar POS, a teen's activity sign-up, a crew tablet handoff, or a guest's own phone at home. Extending it means buying, installing, and maintaining capture hardware and matcher capacity at every touchpoint across 90+ vessels — and it makes Carnival the long-term custodian of a central biometric database, with all the breach exposure and retention obligations that carries.

One more gap sits before the border itself: **document readiness.** Denied boardings and terminal exceptions are mostly a document-quality problem, and document quality is fixed at home, not at the pier. Ultrapass verifies ICAO Doc 9303

machine-readable travel documents and U.S. driver's licenses against authoritative state DMV records (AAMVA DLDV) on the guest's own device at booking or online check-in — surfacing expired, damaged, or non-compliant documents weeks before the terminal instead of at the gangway. Terminal document readers remain fully complementary; Ultrapass is software and displaces none of them.

Ultrapass inverts the model: the identity engine ships as software to the devices already standing at every touchpoint.

4. The Ultrapass Difference: Answers Travel, Biometrics Don't

The unit of value in Ultrapass is the **signed verdict**: a cryptographically verifiable assertion carrying exactly the answers a business decision needs — identity-match result, liveness result, age-over-threshold, credential attributes, and the assurance level that actually ran. Verdicts are produced on the device and delivered over standard federation (OIDC claims, SAML attributes) into whichever Carnival application asked the question. A point of sale, the HUB app, a door controller, or the bridge's muster dashboard receives an answer — never a biometric. Every verdict is auditable and revocable; the matcher itself is a sealed, on-device signal source.

For the engineering and product teams, four properties fall directly out of that design:

- **Kilobytes, not images.** A server-side matching architecture uploads hundreds of kilobytes to megabytes of imagery per transaction; an Ultrapass transaction is a kilobyte-scale signed verdict. On a vessel where thousands of guests share satellite capacity, identity traffic becomes rounding error instead of a line item.
- **Offline by design.** Enrollment and credential issuance need connectivity; verification does not. Matching and liveness run on device, verdicts are signed locally, and signed events queue and sync when links return. The identity system that matters most in an emergency is the one still working when the network is not.
- **No central biometric database — ever.** There is no gallery to secure, no retention schedule to administer, no breach-response plan for a biometric store, because none exists. This settles the GDPR Article 9 / BIPA-class exposure architecturally rather than contractually.
- **Standards in, standards out — the end of SDK lock-in.** Carnival reports that many current applications are coupled directly to the incumbent matcher SDK. Ultrapass ends that pattern instead of repeating it: applications bind to OIDC/SAML federation and standard credential formats (ISO/IEC 18013-5 mobile documents, W3C Verifiable Credentials, OID4VCI/OID4VP), and the matcher becomes an implementation detail behind the verdict interface. If Carnival ever changes biometric engines again, the applications don't change with it.

One engine, four assurance tiers

A single ceremony engine produces one verdict shape across four assurance tiers; each verdict records which tier actually ran, so applications gate policy by assurance level without four separate integrations:

Tier	Mechanism	Offline behavior	Typical Carnival use
1	Platform FIDO2 authenticator; hardware-bound key on the device	Fully offline-capable	Crew single sign-on on managed devices; guest app login
2	On-device biometric verification over the camera — no FIDO hardware dependency	Verification, including age, completes on device	Shared shipboard terminals, POS age checks, older guest devices — the decisive tier for a mixed device fleet
3	Cross-device FIDO hybrid; the phone authenticates the android phone or tablet session, bound by Bluetooth LE proximity	Requires local BLE link	Embarkation android phone, dining check-in stands or tablets

Tier	Mechanism	Offline behavior	Typical Carnival use
4	Pure-web browser capture (contractually scoped deployments)	Online	Pre-cruise Online Check-In with no app install

5. The Right Tool: Every Shipboard Question Is a 1:1 Question

Carnival's team raised face-recognition accuracy as a discriminator. The honest answer starts with a category distinction, because a gallery-search platform and an on-device verifier do not solve the same problem. A **1:N identification** engine answers "who, among N enrolled faces, is this?" — a search against a central gallery whose false-match risk compounds with gallery size, which is why watchlist deployments are engineered around candidate lists and human adjudication. That is what Thales Cogent-class engines are built for, and it is the right architecture for government watchlist screening. A **1:1 verification** engine answers "is this the person who enrolled?" — one live, liveness-gated capture against one protected template on the user's own device, the same match-on-device paradigm consumers exercise every time they unlock a modern smartphone. No gallery, no compounding search error, no adjudication queue.

The most important question, is this the enrolled guest charging to this folio? Is this guest over 21 at the casino entrance? Is this crew member at their assigned muster station? Is this the same passenger re-boarding at the gangway? **Every one is a 1:1 question.** Procuring a 1:N search engine to answer them imports the 1:N error model — and the central gallery that comes with it — into use cases that need neither. The one genuinely gallery-based shipboard function, associating photographer images with passengers in the HUB app, can remain on the incumbent engine unchanged for as long as Carnival finds it valuable (Section 10).

Private Identity's verifier is measured where 1:1 verifiers are measured: NIST's Face Recognition Technology Evaluation (FRTE) 1:1 Verification track, as **privid_004** — with a public report card covering false non-match rates at fixed false match rates across operational datasets, demographic differential analyses, face-mask effects, and template aging. (Per NIST policy, FRTE results reflect participation and measured performance, not NIST endorsement.) And to be precise about the incumbent: server-side 1:1 verification exists in the Thales product family too — so the differentiator is not the arithmetic but the architecture. A server-side 1:1 match still uploads imagery per transaction, still maintains a central template store, and still fails with the link. Ultrapass performs the 1:1 match on the device, behind certified liveness, bound to FIDO2 keys, with nothing biometric on the wire.

An on-device verifier has no central gallery to search, to secure, or to mis-match against — one architectural fact that settles the privacy question and the accuracy question together.

6. Two Products, One Ultrapass SDK

6.1 Carnival DigitalID — for guests

Delivered inside the existing Carnival HUB app and pre-cruise Online Check-In. Full journey in Appendix A.

- **Enroll at home:** selfie + passport (ICAO Doc 9303), or U.S. driver's license verified against state DMV records, producing remote identity proofing to NIST 800-63 IAL2 and issuing a verifiable credential into the guest's Carnival DigitalID wallet — the first digital identity wallet in the cruise industry, holding verified identity, age predicates, folio linkage, and loyalty tier.
- **Carry a verified credential:** presented as an Ultrapass QR with data mapping — each scan point receives exactly the attributes Carnival maps to it (an over-21 predicate at the casino entrance, folio ID at the Fun Shops register)

and nothing more. Selective disclosure enforced by design, not by policy memo. Also presentable over Bluetooth for proximity flows; fully functional offline at sea.

- **Selfie re-authentication:** on-device face match + FIDO2 passkey (AAL2) approves purchases, bookings, account changes, and card-on-file transactions — with a signed audit trail of the verified individual behind every action.
- **Real-time age proof:** from the verified credential, or live on-device facial age estimation returning only an over/under-threshold answer — supporting alcohol, casino (jurisdiction-configurable 18/21 gating), and adult-only experiences on guest phones or crew tablets.
- **Family accounts:** parents approve minors' activities and purchases with a selfie from anywhere on the ship — turning parental control into a spend enabler rather than a spend blocker.

6.2 Carnival CrewID — for the workforce

Delivered on shared ship devices, federated with TmHub and shipboard systems. Full journey in Appendix B.

- **Fast crew enrollment:** document verification at hire, identities pre-loaded to ship devices, face + passkey setup in minutes — built for the industry's highest-tempo turnaround days and a large multinational workforce.
- **Shared-device login:** any crew member, any authorized device — sub-second on-device face match + passkey, no PINs, passwords, or card swipes. Eliminates credential sharing, the single largest hole in shipboard accountability.
- **One login for every app:** a single Face-ID-like login federated across dining, guest services, POS, housekeeping, inventory, property-management, and HR systems, with fast session handoff as crew rotate devices.
- **Manager selfie approvals:** comps, refunds, and voids approved in seconds on the nearest tablet, with cryptographic proof of who approved what — built for high-turnover dining environments.
- **Bluetooth presence, signed:** BLE-ranged presence events enter the verdict stream — muster-station accountability, watchkeeping presence, restricted-zone entry — without deploying dedicated badging hardware.
- **Time-in/Time-out and Training:** Unique identity authentication for crew members allowing secure and seamless time-in and time out along with ensuring the crew member completed the training themselves.

7. The Journeys: From the Couch to the Casino to the Muster Deck

Pre-cruise

At Online Check-In, the guest verifies from home — in the browser or the HUB app: document verification, live face capture with liveness, age predicates, wallet issuance. Document problems surface weeks early, when they are fixable, instead of at the pier as a denied boarding. Guests who arrive verified board faster and start spending sooner — on day one of a 7-day cruise, an hour recovered is bar, casino, and excursion-desk time.

Embarkation day and Gangway

Wallet-ready guests move through existing terminal processes unchanged; where Carnival deploys kiosks, the guest's phone authenticates the kiosk session over Bluetooth proximity. The industry benchmark for what verified-before-arrival unlocks: ~94% reduction in per-passenger processing time and total embarkation cut from ~2 hours to ~30 minutes.

Onboard commerce

Dining venues recognize the guest and their entitlements from a QR presentation or the guest's own device — no camera at the podium. Bars and the Fun Shops receive a signed over-threshold age predicate, eliminating document handling at the register and unlocking self-service ordering channels currently blocked by the age-check problem.

Selfie-approved card-on-file purchases cut abandoned transactions and friendly-fraud chargebacks, with an IAL2-verified, AAL2-authenticated individual behind every high-value transaction.

Safety and emergency procedures

Carnival's self-mustering safety briefing — in place since July 2021 and compliant with SOLAS Chapter III, Regulations 19.2.2 and 19.2.3 — has guests check in with muster-station teams post embarkation. Ultrapass upgrades that check-in to a verified, signed event: Bluetooth presence at the assigned station, QR fallback, crew-tablet override, rolling up to a real-time accountability view on the bridge. Crew accountability at emergency stations resolves in seconds of signed attestation rather than minutes of radio roll-call — and the whole flow **degrades gracefully through network loss,** continuing on-device and synchronizing when links return.

Ports and debarkation

U.S. terminal debarkation via CBP TVS is untouched. At foreign ports, gangway re-boarding checks run on crew devices, offline-capable, against the guest's wallet credential — supplementing the existing voyage facial-recognition process, not replacing it.

Crew operations

One federated identity across shipboard systems, verified documents at hire and faster crew changes, proximity-verified watchkeeping and restricted-zone access — all issued into the same wallet, all emitting the same verdict shape, all recoverable as thousands of labor-hours per day across ~100,000 shipboard employees.

8. The Money: Illustrative Financial Impact

Figures are directional and deliberately conservative, with assumptions stated inline; they can be refined jointly with Carnival's own operational data. All projected benefits are estimates based on publicly available data.

Value driver (assumption)	Mechanism	Annual benefit (base case)
Onboard revenue lift of 0.25–0.5% on ~$12B onboard/other revenue (faster transactions, age-gated channels unlocked, family accounts, earlier day-one boarding)	More completed transactions per guest-day	**$30–60M**
Workforce time recapture: 3 min/day × 100,000 shipboard crew × 350 days × ~$15/hr loaded cost	Labor capacity reallocated to guest service (not headcount reduction)	**~$26M equivalent capacity**
Terminal & check-in efficiency: reduced agent-minutes per guest and shorter turnaround critical path (benchmark: ~94% processing-time reduction at biometric embarkation)	Port labor + earlier onboard spend window	**$10–25M**
Fraud, chargebacks, and shrinkage: a verified individual behind every card-on-file transaction, comp, void, and refund	Loss reduction + dispute win-rate	**$10–40M**
Avoided hardware TCO for any expansion of biometric touchpoints: venue capture cameras, install, spares, GPU matcher capacity, central biometric infrastructure and its security	CapEx + OpEx avoidance vs. hardware-anchored expansion	**Deployment-dependent; material**

Value driver (assumption)	Mechanism	Annual benefit (base case)
burden, satellite bandwidth, and biometric-privacy litigation exposure		
Indicative total		**$75–150M+ / year**

The cost side is what makes the ROI durable: Ultrapass is licensed as software — per-user/per-ship SDK licensing plus integration services. No capture-hardware CapEx, no shipboard install program, no matcher farm, no middleware estate. Even at the conservative end of the benefit range, payback is measured in months, because the denominator is small.

9. Proof, Standards, and Compliance

The engine is not a pilot-stage promise. It runs in production today across the globe providing millions of **on-device biometric verifications and age estimations per month** for large enterprises — with active deployments and engagements across regulated enterprises from national retail and pharmacy chains to casinos and and financial services companies.

Domain	Standard / program	Claim
Authentication standards	FIDO2 / WebAuthn / CTAP2	Standards-conformant implementation; FIDO Alliance voting member; FIDO certification targeted Q4 2026 (not yet certified)
Cryptography	NIST CMVP	Validated cryptographic module — certificate #4985
Presentation-attack detection	ISO/IEC 30107-3	ACCS 1:2020 — Level 3 (Highly Effective); every capture is liveness-gated — a stored photo, screen replay, or mask does not enroll and does not verify
Face recognition evaluation	NIST FRTE 1:1 Verification	Participant — privid_004 (participation and measured performance; not a NIST endorsement)
Biometric template protection	ISO/IEC 24745; ISO/IEC 30136	Protected, irreversible, cancelable on-device templates
Identity assurance	NIST SP 800-63	IAL2 identity proofing / AAL2 authentication (Kantara scope)
Credential formats	ISO/IEC 18013-5; W3C VC; OID4VCI / OID4VP	Standards-based wallet issuance and presentation — no proprietary credential format
Privacy posture	GDPR; CCPA; BIPA-class statutes	On-device processing; no biometric image or template leaves the device; verdict-only transport

The regulatory direction of travel favors this architecture. The March 2026 AEPD decision against a server-side age-assurance provider — €950,000 across unlawful Article 9 biometric processing, invalid consent, and excessive retention, currently under appeal — attaches in every particular to central processing and central retention. Carnival

already deletes voyage facial-recognition data post-cruise; **Ultrapass simply never collects it centrally in the first place.**

10. Coexistence — and the Exit from Lock-In

Nothing existing is removed at any stage. Ultrapass coexists on day one with the Thales-based voyage capability and the CBP TVS partnership, touching neither, and it never sits in the critical path of a process that already works. **There is no rip-and-replace anywhere in this proposal.**

Where Carnival's applications are today coupled directly to the incumbent matcher SDK, the verdict layer provides a low-risk exit: an application adds consumption of signed verdicts over OIDC/SAML — roughly two weeks of work — while its existing integrated code paths keep running. Traffic shifts flow by flow, and legacy SDK calls are decommissioned per application, on Carnival's schedule, only after the verdict path has proven itself in production. No big-bang cutover exists in this model.

Function	Nature	Disposition
Authentication, age predicates, venue access, muster accountability, commerce, gangway re-boarding	1:1 verification	Migrates to on-device Ultrapass verdicts, flow by flow, at Carnival's pace
Voyage photo association — photographer images matched to passengers in the HUB app	Voyage-scoped 1:N gallery association	May remain on the incumbent engine, unchanged, for as long as Carnival finds it valuable
U.S. terminal debarkation	Government biometric infrastructure (CBP TVS)	Untouched throughout

And because applications bind to open standards rather than a vendor API, the deepest cost of the current position — application-level coupling to a specific matcher — ends here. **That is what "better than Thales" means at the portfolio level: not a better gallery, but the end of gallery coupling. This is the last identity migration of its kind.**

11. Why Adoption Cost Is Low

The rollout plan is Carnival's call. What the architecture guarantees is that the cost and risk of adopting Ultrapass are structurally low compared to any hardware-based expansion:

- **~1-2 weeks of integration per business application.** Native SDKs for iOS, Android, and web plus standards federation slot into pre-cruise Online Check-In, the Carnival HUB app, Funtime, and TmHub without re-architecture; integration patterns are already documented for twelve identity-provider and orchestration platforms. (**POC and production launch is dependent on Carnival Cruise engineering cycles**)
- **No middleware estate.** There is no Ultrapass middleware tier to procure, size, harden, or patch across 90+ vessels — the heavy computation has already happened on the device before the network is involved at all.
- **Software distribution, not shipyard schedules.** Deployment rides app-store and MDM channels; rollout velocity is decoupled from drydock calendars, terminal construction, and port infrastructure projects.
- **Scales at near-zero marginal cost per touchpoint.** Adding a venue, device, or use case is a configuration change, not a purchase order. Model updates ship inside ordinary app releases.
- **Provable in one sprint.** A single two-week integration against pre-cruise Online Check-In in a Carnival test tenant demonstrates the entire thesis — verified guests, signed verdicts, zero new infrastructure — before any commercial commitment.

12. Risks and Mitigations

Risk	Assessment	Mitigation
Guest adoption of enrollment	Enrollment friction could cap benefits	Incentivize (priority boarding lane, onboard credit); opt-in with manual fallback preserved
Biometric privacy regulation (BIPA, GDPR, state laws)	Central biometric stores are the litigation magnet	The architecture avoids the risk class: on-device biometrics, no central biometric database, selective disclosure, explicit consent flows
Crew acceptance / labor relations	Perception of surveillance	Position as friction removal (no PINs/cards) and fair attribution; on-device matching, transparent policy, works councils engaged early
Integration complexity with legacy shipboard systems	POS/PMS heterogeneity across brands	SDK + standards-federation pattern with a ~1-2-week-per-app track record; scoping can start with the hardest integrations first
Vendor concentration	Adding a second identity vendor	Standards-based (NIST 800-63, FIDO2, ISO 18013-5, W3C VC) — credentials and passkeys are portable; coexists with Thales rather than forcing migration; ends matcher-SDK coupling rather than repeating it
Offline edge cases	Connectivity loss mid-voyage	Offline-native design is a core property of the architecture, not an afterthought — verification, liveness, and event signing continue through network loss

13. What Only Private Identity Brings

Thales secures the gangway. Ultrapass monetizes everything after it. The incumbent model is dedicated hardware and central matching at fixed control points; Ultrapass is a software-only, edge-native, offline-capable identity layer that reaches every guest phone, crew tablet, POS terminal, and kiosk Carnival already owns — and every identity moment those devices touch.

In one SDK, and only from Private Identity:

- **The industry's first guest-held digital identity wallet** — standards-based (ISO/IEC 18013-5, W3C VC), issued from remote NIST 800-63 IAL2 proofing completed at home, presented as QR with selective data mapping or over Bluetooth, fully functional offline at sea.
- **Answers on the wire, never biometrics** — kilobyte-scale signed verdicts into Carnival's existing applications over OIDC/SAML, with no central biometric database and no satellite dependency in the identity critical path.
- **AAL2 passwordless authentication everywhere** — on-device face + FIDO2 passkey for guest transactions and 100,000 crew on shared devices, with certified Level 3 liveness on every event.
- **New revenue-bearing use cases** — real-time age assurance, selfie-approved transactions, family accounts with parental sign-off, manager selfie approvals, and signed muster accountability — none of which a hardware-anchored model can economically serve.

- **A cost structure that makes the ROI obvious** — no capture hardware, no matcher farm, no middleware estate, ~2-week integrations, and an illustrative base case of $75–150M+ per year.
- **Zero disruption — and a permanent exit from lock-in** — CBP and voyage facial flows continue exactly as they are, while Carnival's applications decouple from any matcher SDK for good.

Carnival led the industry to 100 million guests. The next competitive frontier is not a bigger ship — it is removing every remaining second between a guest's intent and Carnival's revenue, and between a crew member and their work. **Ultrapass makes Carnival the first cruise line to put a high-assurance digital identity in every guest's pocket and a passwordless verified login in every crew member's face — with software the fleet already has the hardware to run.**

Verified guests. Verified crew. Faster service. Better control.

Appendix A — Carnival DigitalID (Guest Journey)



Carnival DigitalID: enrollment at home, verified credential, selfie re-authentication, real-time age proof — privacy-preserving by design.

Appendix B — Carnival CrewID (Workforce Journey)



Carnival CrewID: fast enrollment, shared-device face + passkey login, one login for every app, manager selfie approvals.

Appendix C — Sources

[1] Carnival Cruise Line, "Facial Recognition Technology" (help.carnival.com). [2] U.S. DHS/CBP, Privacy Impact Assessment DHS/CBP/PIA-056, Traveler Verification Service. [3] Thales, "Cogent Face Recognition Platform" brochure (May 2020). [4] Thales/TSSL case study, Gemalto Document Reader AT9000 MK2 — "major cruise line" (customer unnamed). [5] Carnival Cruise Line, "Safety Briefing — Muster Station Drill" (self-mustering since July 2021; SOLAS Ch. III Reg. 19.2.2–19.2.3). [6] AEPD resolution re Yoti Ltd, €950,000, published March 10, 2026; under appeal. [7] NIST FRTE 1:1 Verification report card — Private Identity LLC, algorithm privid_004 (submitted March 5, 2025; report card generated March 29, 2026). Market benchmarks: Carnival Corporation & plc FY2025 results (revenue $36.4B; onboard & other ≈ one-third of revenue; ~100,000 average shipboard employees; ~225,000 daily guests); U.S. CBP seaport biometrics program (debarkation clearance −30%, ~2-second verification, >98% accuracy); Port of Vancouver biometric embarkation (~2–3 min → ~10 s per passenger; ~2 h → ~30 min total). Financial figures in Section 8 are Private Identity illustrative estimates, not Carnival guidance. Thales attribution per Carnival working session, July 15, 2026; public sources do not identify the vendor.